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$$I = \int e^x \cos(x) dx$$

$$\begin{cases} u = \cos(x) \\ dv = e^x dx \end{cases} \Rightarrow \begin{cases} du = -\sin(x) dx \\ v = e^x \end{cases}$$

$$I = e^x \cos(x) + \int e^x \sin(x) dx$$

$$\begin{cases} t = \sin(x) \\ dw = e^x dx \end{cases} \Rightarrow \begin{cases} dt = \cos(x) dx \\ w = e^x \end{cases}$$

$$I = e^x \cos(x) + e^x \sin(x) - \int e^x \cos(x) dx$$

$$2I = e^x \cos(x) + e^x \sin(x) + c$$

$$I = \frac{e^x (\cos(x) + \sin(x))}{2} + c$$

References